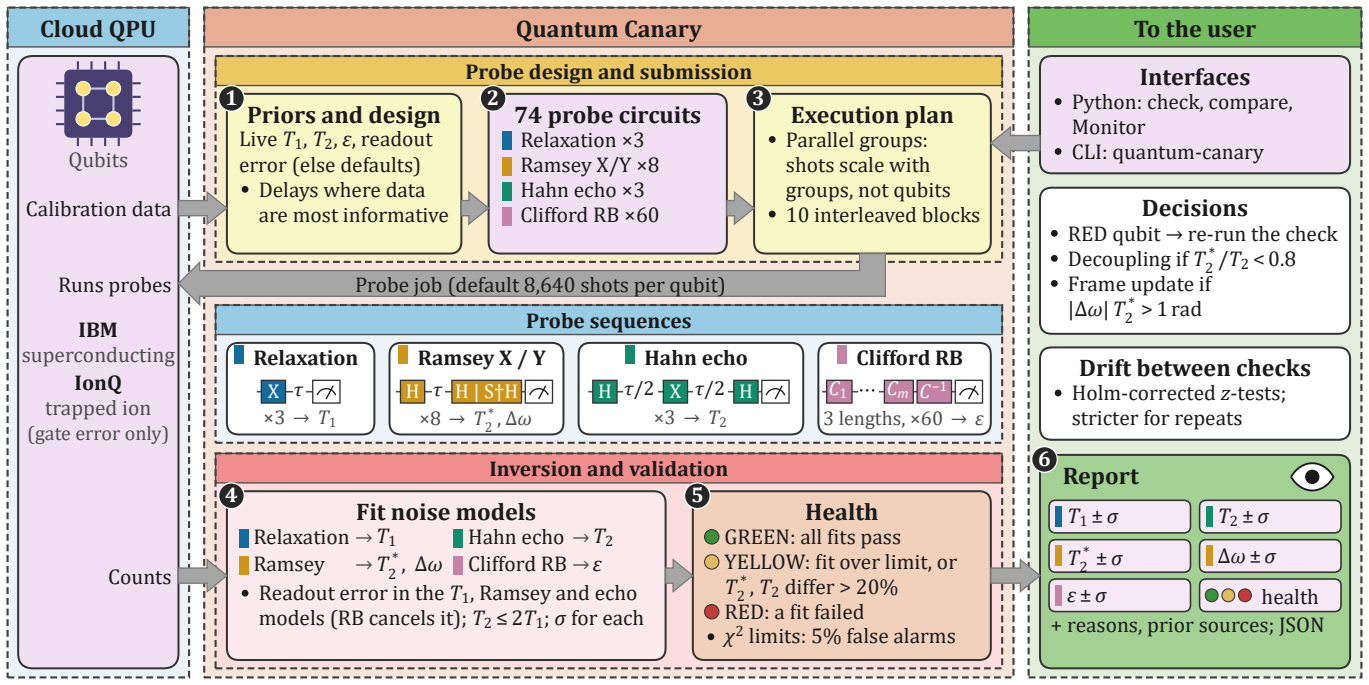

Cloud QPU
Qubits
Calibration data
Runs probes
IBM
superconducting
IonQ
trapped ion
(gate error only)
Counts
Quantum Canary
Probe design and submission
1
Priors and design
Live $T_1$, $T_2$, $\varepsilon$, readout error (else defaults)
• Delays where data are most informative
2
74 probe circuits
Relaxation ×3
Ramsey X/Y ×8
Hahn echo ×3
Clifford RB ×60
3
Execution plan
• Parallel groups: shots scale with groups, not qubits
• 10 interleaved blocks
Probe job (default 8,640 shots per qubit)
Probe sequences
Relaxation
X – τ
×3 → $T_1$
Ramsey X / Y
H – τ – H | S†H
×8 → $T_2^*$, $\Delta\omega$
Hahn echo
H – τ/2 – X – τ/2 – H
×3 → $T_2$
Clifford RB
$C_1$ ··· $C_m$ $C^{-1}$
3 lengths, ×60 → $\varepsilon$
Inversion and validation
4
Fit noise models
Relaxation → $T_1$
Hahn echo → $T_2$
Ramsey → $T_2^*$, $\Delta\omega$
Clifford RB → $\varepsilon$
• Readout error in the $T_1$, Ramsey and echo models (RB cancels it); $T_2 \le 2T_1$; $\sigma$ for each
5
Health
GREEN: all fits pass
YELLOW: fit over limit, or $T_2^*$, $T_2$ differ > 20%
RED: a fit failed
• $\chi^2$ limits: 5% false alarms
To the user
Interfaces
• Python: check, compare, Monitor
• CLI: quantum-canary
Decisions
• RED qubit → re-run the check
• Decoupling if $T_2^*/T_2 < 0.8$
• Frame update if $|\Delta\omega|\, T_2^* > 1$ rad
Drift between checks
• Holm-corrected z-tests; stricter for repeats
6
Report
$T_1 \pm \sigma$
$T_2 \pm \sigma$
$T_2^* \pm \sigma$
$\Delta\omega \pm \sigma$
$\varepsilon \pm \sigma$
health
+ reasons, prior sources; JSON